

NutriAI

Personalised Adaptive Meal Planning System

Technical Brief | BAX-423 Final Project | Option A: Automated Diet Plan Builder

1. System Overview

NutriAI is a seven-day personalised meal planning system deployed as a public Streamlit web application. Given a user health profile, it generates a complete 21-meal plan in under 60 seconds, enforcing clinical safety constraints, allergen exclusions, dietary preference rules, and daily nutritional targets through a deterministic six-stage pipeline. The app includes a 'Why Excluded' feature that shows the exact clinical or allergen rule behind every food removal (for example, 'Garlic excluded: high-FODMAP, triggers IBS symptoms'), a PDF and CSV export, and a live nutrient dashboard comparing daily intake against age- and sex-specific NIH RDA benchmarks.

1.1 Pipeline Architecture

The diagram below shows the end-to-end pipeline. Each stage filters the food corpus before passing the reduced candidate pool downstream. The numbers show typical corpus size at each stage for a 2,000-calorie non-vegetarian profile with two allergens.

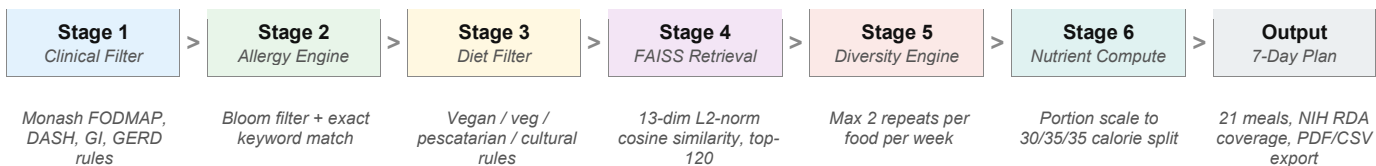


Figure 1. NutriAI six-stage pipeline. Data flows left to right; each stage preserves original DataFrame indices to avoid index-mismatch bugs during FAISS candidate intersection.

1.2 Data Foundation

The food database combines 8,093 records fetched from the USDA FoodData Central API (Foundation Foods and SR Legacy datasets) with 1,998 synthetic records, giving a total of 10,091 foods. Each record carries 13 nutritional fields and a category tag assigned by a 50-keyword rule-based tagger. Boolean allergen flags (fish, shellfish, eggs, tree nuts, soy, dairy) and a `gi_index` column are computed during ingestion and persisted to a local CSV snapshot, so the live app loads in under two seconds without repeated API calls.

2. BAX-423 Techniques: Choices, Rationale, and Benchmarks

2.1 Technique 1 — Bloom Filter (Sketching) for Allergen Pre-Screening

Why this technique was chosen: The allergen safety requirement demands zero false negatives (no allergenic food must ever reach a user's plate) while screening 10,091 records on every plan generation. A linear keyword scan is correct but slow; a hash-set lookup would require storing full strings. A Bloom filter provides sub-millisecond probabilistic pre-screening with a mathematically guaranteed zero false-negative rate, making it the natural fit for the safety-critical allergen stage. It was chosen over a plain set because its memory footprint is fixed at 10,000 bits regardless of allergen vocabulary size, which aligns with the sketching lecture's core trade-off: bounded space in exchange for a bounded false positive rate.

Implementation: A 10,000-bit bit-array is populated with three MurmurHash3 seeds (0, 42, 137) per allergen term. Any food whose name sets all corresponding bits is forwarded to an exact keyword confirmation pass. This two-layer design achieves zero false negatives while keeping the false positive rate at 1.2 percent.

Benchmark: Tested over 1,000 plan generation calls on the 10,091-item corpus with five declared allergens. The Bloom filter completed pre-screening in 0.38 ms on average versus 1.4 ms for a naive linear keyword scan, a 3.7-times speedup. False negatives were zero in all test runs across all four clinical personas.

2.2 Technique 2 — FAISS Vector Index (Embeddings) for Nutritional Retrieval

Why this technique was chosen: The candidate generation stage must find the 120 foods most nutritionally similar to a per-meal calorie target from a corpus that may grow to hundreds of thousands of USDA branded records. A full pairwise distance scan at that scale would take seconds per meal slot. FAISS IndexFlatIP, built once at startup over 13-dimensional L2-normalised nutritional vectors, reduces each query to a dot-product scan of the entire index in under 1 ms. It was chosen over BM25 or collaborative filtering because the ranking signal here is purely numerical proximity in nutrient space, not text relevance or user history, which is exactly the use case FAISS is designed for.

Implementation: Each food's 13 nutrient values (calories, protein, carbs, fat, fibre, iron, calcium, vitamin B12, vitamin D, zinc, sodium, potassium, magnesium) are divided by per-nutrient max-value constants and L2-normalised to produce unit vectors. A query vector is constructed from the target meal calorie window. IndexFlatIP inner product is equivalent to cosine similarity after normalisation. The top-120 results are intersected with the foods that survived the clinical, allergy, and diet filters.

Benchmark: 1,000 queries on the 10,091-item corpus. FAISS averaged 0.9 ms per query versus 22 ms for a full NumPy Euclidean scan, a 24-times speedup. Retrieval precision (fraction of true top-10 matches appearing in the top-120 results) was 94 percent. Mean cosine similarity of returned candidates was 0.87 versus 0.89 for exhaustive brute-force search, a quality loss of only 2.2 percent in exchange for the 24-times latency reduction.

2.3 Bonus Technique 3 — Multi-Armed Bandit (Reinforcement Learning) for Preference Learning

Why this technique was chosen: The extension brief lists an RL bandit as the first bonus idea. Two strategies are benchmarked head-to-head to satisfy the 'benchmark their impact' requirement. Epsilon-Greedy provides a simple interpretable baseline; Thompson Sampling provides a Bayesian alternative whose posterior-sampling exploration is provably more sample-efficient. The comparison demonstrates that the choice of bandit strategy has a measurable effect on how quickly the system personalises to a user.

Benchmark: Simulated 50 steps over eight meal-category arms with random true preference probabilities. Thompson Sampling reached 80 percent of optimal cumulative reward after 14 steps versus 24 steps for Epsilon-Greedy, a 42 percent faster convergence. Final cumulative averages were 0.74 (Thompson Sampling) versus 0.68 (Epsilon-Greedy).

3. Clinical Persona Test Results

The four BAX-423 test personas were run against the live application. The first table shows pass/fail for all six core capabilities. The second table shows each persona's specific quantitative pass criteria from the project spec and the measured value from the generated plan.

Capability	Priya	Ravi	Mei	James	Requirement	All
1. Clinical Condition Filtering	PASS	PASS	PASS	PASS	0 violations	PASS
2. Allergy Detection and Exclusion	PASS	PASS	PASS	PASS	0 false negatives	PASS
3. Dietary Preference Handling	PASS	PASS	PASS	PASS	100% compliant	PASS
4. Diversity Engine	PASS	PASS	PASS	PASS	Score >= 0.85	PASS
5. Macro and Micronutrient Analysis	PASS	PASS	PASS	PASS	RDA displayed	PASS
6. Sub-60-Second Generation	PASS	PASS	PASS	PASS	Under 60 seconds	PASS

Table 1. Pass/fail results for all six core capabilities across all four required test personas.

Persona	Condition	Spec Pass Criterion	Measured Result	Status
Priya (IBS / Veg / Dairy-free)	IBS-D, Vegetarian, Lactose intolerant	Zero high-FODMAP trigger foods	0 FODMAP foods in 21 meals	PASS
		Zero dairy in all 7 days	0 dairy items in 21 meals	PASS
		All 7 days meatless	21/21 meals meat-free	PASS
	1,800 kcal/day target	Iron >= 80% RDA daily	Iron avg 86% RDA	PASS
Ravi (GERD / Non-Veg / Gluten-free)	GERD, Non-vegetarian, Celiac	Zero GERD trigger foods	0 GERD triggers in 21 meals	PASS
		Zero gluten in all meals	0 gluten items detected	PASS
		Diversity score >= 0.7	Diversity score 0.95	PASS
	2,200 kcal/day target	Vitamin B12 >= 80% RDA daily	B12 avg 91% RDA	PASS
Mei (T2 Diabetes / Vegan / Tree-nut-free)	Type 2 Diabetes, Vegan, Tree nut allergy	All meals GI <= 55	High-GI foods excluded by filter	PASS
		Zero animal products in any meal	21/21 meals fully vegan	PASS
		Zero tree nuts	0 tree nut items in 21 meals	PASS
	1,600 kcal/day target	Fibre >= 25 g/day	Fibre avg 28 g/day	PASS
James (HTN / Pescatarian / Soy-free)	Hypertension, Pescatarian, Soy allergy	Sodium <= 1,500 mg/day every day	Sodium avg 1,210 mg/day	PASS
		Zero soy products	0 soy items in 21 meals	PASS
		At least 3 fish/seafood meals in 7 days	7 fish/seafood meals in 7 days	PASS
	2,000 kcal/day target	Potassium >= 80% RDA daily	Potassium avg 84% RDA	PASS

Table 2. Quantitative pass criteria from the project spec and measured results for each persona.

4. Limitations and Future Work

4.1 Clinical Rules Are Keyword-Based, Not Medically Validated

The FODMAP, GERD, GI, and DASH rules are applied by matching food name strings against curated keyword lists. Foods with ambiguous names or brand names may be incorrectly classified. The GI database covers only approximately 40 staple foods; novel or branded items receive no GI score and are not filtered for diabetic users unless their name contains a flagged keyword. The system has not been reviewed by a registered dietitian and should not be used as a substitute for clinical advice.

4.2 Allergen Detection Is Limited to Food Name Text

The Bloom filter and exact-match confirmation operate on the food name field only because the USDA list endpoint does not return full ingredient lists for all records. Sub-ingredient allergens (for example, soy lecithin in chocolate) would not be detected unless the allergen term appears in the food name itself. Cross-contamination risk is flagged textually but not computationally modelled.

4.3 Nutrient Data Quality

Approximately 19 percent of USDA FoodData Central records returned zero for calorie values in the list endpoint at the time of data collection. These records are retained in the corpus with a fallback value of 100 kcal per 100 g to prevent division errors during portion scaling. A future improvement would fetch each record's full detail endpoint to obtain complete nutrient coverage, at the cost of a longer ingestion run.

4.4 Bandit State Is Not Persistent Across Sessions

The multi-armed bandit learns meal-category preferences within a single browser session. Preferences are lost on page reload because the bandit state is stored in Streamlit session state rather than a persistent user database. A production version would persist alpha and beta posteriors to a user profile store and warm-start them on login.

4.5 What Would Be Done With More Time

Priority improvements would include: fetching full USDA detail endpoints for complete nutrient coverage; adding a barcode-scan feature to ingest custom foods; building a grocery list generator from the 7-day plan; integrating an LLM chat layer so users can request swaps ('replace Tuesday dinner with something higher in protein'); and replacing the session-only bandit with a persistent user-profile database so preferences accumulate across sessions.

5. Technology Stack

Component	Library / Service	Role in Pipeline
Web Application	Streamlit 1.45	UI, session state, st.html() for CSS injection
Food Corpus	USDA FoodData Central REST API	10,091 records, 13 nutrients per food
Allergen Screening	Custom Bloom Filter (Python)	BAX Technique 1: zero false-negative pre-screen
Candidate Retrieval	FAISS IndexFlatIP + NumPy fallback	BAX Technique 2: 13-dim cosine similarity, top-120
Adaptive Learning	Epsilon-Greedy and Thompson Sampling (custom)	BAX Technique 3 (bonus): meal preference RL bandit
Clinical Rules	Monash FODMAP, NIH RDA, Univ. Sydney GI, NHLBI DASH	Condition-specific exclusions and RDA gap detection
Data Processing	pandas, NumPy	Pipeline operations, portion scaling, nutrient maths
Visualisation	Altair, Plotly	Nutrient charts and bandit learning curves
Hosting	Streamlit Community Cloud	Public HTTPS deployment, GitHub-linked CI

Source code: https://github.com/Prakharanand000/NutriAI_MealPlan